DAX - Data Analysis Expression

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DAX: Data Analysis Expression is a formula expression language used in power BI

DAX formula includes functions, operators and values to perform advance calculations.

Two way to calculate DAX

→ calculate column

→ Measure

Calculated column: values are calculated based on information each row of table → it will create new physical column in table.

Measure: values are calculated based on information from any filters on the report (filter context) → on fly it will calculate the values

Syntax:

Measure Name   DAX function   Table Name

Total Sales = SUM(MasterSales'[Sales])   → column Name

DAX operators:

Arithmatic operator = +, −, *, /

Comparison operators: =, >, >, >=, <=, <>,

logical operators = && → AND

|| → OR

IN → In

DAX function categories:

Logical function   Text   Filter functions   Date

DAX function categories.

Math & Stats function	logical function	Text function	Filter function	Date & Time function
Sum	IF	Concatenate	calculate	Date
Average	Switch	Combine	Filter	Date Diff
Count	IF-Error	Format	ALL	Today/Now
COUNTA	AND	Left/Mid/Right	ALLExcept	DatesYTD
MIN	OR		AllSelected	DatesMTD
MAX				DateADD
DistinctCOUNT				

// Stats & Math function:

SUM → Total Sales = SUM('Master sales'[sales])

→ Evalue the Sum of column → SUM(columnName)

Average → Return the Average of All the Number in the column

= Average(column Name)

COUNT = Count the Number of NonEmpty cells in the column → Excluding Boolean values

= COUNT (columnName)

COUNTA = count the Number of non-Empty cell in the Column including Boolean values

= COUNTA (column Name)

DistinctCount = Count the number of distinct values in column

= DISTINCTColumn(columnname)

MIN/MAX = Find min/Max value in the column

= Min(columnName)/MAX(column Name)

Logical function:

IF : Check if given condition is met and returns one value if the condition is True and another if condition is FALSE

= if (Logical Test , Result True , Result false)

Ex = if (Sales >= 25 "High sales", "Low Sales")

IfError : Evaluates an expression and return a specified value if it returns an Error. otherwise it returns the Expression itself.

= IFError (value, valueError)

Switch : Evaluates the expression against a list of values and returns one of multiple possible expressions.

= Switch (Expression , Value 1, Result 1
" " 2 " 2
" " 3 " 3

Ex Sales category = Switch(True(),
Sales <= 25 "Low Sales",
Sales <= 50 "Medium Sales",
Sales <= 75 "Upper Medium Sales",
"High sales"